Carson Phelan
CSCI 5499A.001
Project 1
Due 10/25/2026

Project 1 Report

# PART 1

## Task 1:

```python
# function to divide image into patches
def patchify(img, p=16):
    """img: (C,H,W) tensor -> (N, C, p, p), non-overlapping, row-major order."""
    C, H, W = img.shape
    patches = img.unfold(1, p, p).unfold(2, p, p)
    return patches.permute(1, 2, 0, 3, 4).reshape(-1, C, p, p)

# show patch information
def show_patches(patches, grid=14):
    fig, axes = plt.subplots(grid, grid, figsize=(8, 8))
    for ax, patch in zip(axes.flat, patches):
        ax.imshow(patch.permute(1, 2, 0).numpy())
        ax.axis("off")
    plt.subplots_adjust(wspace=0.05, hspace=0.05)
    plt.show()
```

Function to Divide Image into Patches.

**Roles of Patch Projection, Positional Embeddings, and the CLS Token:**

Patch Projection is the method by which a vision transformer converts images into tokens. A vision transformer model doesn't directly understand pixel significance, so images are spliced into sections ('tokens') so that the transformer can characterize and quantify the equivalent word embeddings of the selected tokens.

"Positional Embeddings" refers to the way that the model quantifies each patch's positional relation compared to other patches in the image. Each patch is assigned a vector by the model so that each token carries information about where the patch came from in the image. Without this embedding, the overall image completely loses context (an eyebrow shouldn't be below an eye) in a way which prevents the model from accurately predicting contextual understanding.

The CLS token is an extra vector added to the end of each image's sequence (ex. 196 + 1). This token carries the prediction of the model in a single place, and is where the singular predicted value of an image class is stored (i.e. German shepherd (0.88)).

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**My patch visualization vs. vit_b_16's learned patch projection:**

My patch function only rearranges pixels into 196 tiles and does no learning. The model's patch projection is a layer within the transformer that converts each image into a high-dimensional vector mapping, which is what the model actually uses to predict token information.

# Task 2:

Exact Label Mapping of My Images:

| **My Category** | **Exact ImageNet Label** | **ImageNet Index** |
|---|---|---|
| jellyfish | jellyfish | 107 |
| german_shepherd | German shepherd | 235 |
| hummingbird | hummingbird | 94 |
| tarantula | tarantula | 76 |
| tiger | tiger | 292 |

Top Five predictions for Each Image:

| file | TRUE | pred1_idx | pred1 | conf1 | top5 | correct |
|---|---|---|---|---|---|---|
| german_shepherd_1.jpg | german_shepherd | 51 | triceratops | 0.4001022577 | triceratops (0.40); microphone (0.16); corkscrew (0.11); goblet (0.01); strainer (0.01) | FALSE |

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| | | | | | | |
|---|---|---|---|---|---|---|
| german_shepherd_2.jpg | german_shepherd | 235 | German shepherd | 0.8817872405 | German shepherd (0.88); malinois (0.00); briard (0.00); soccer ball (0.00); Norwegian elkhound (0.00) | TRUE |
| german_shepherd_3.jpg | german_shepherd | 235 | German shepherd | 0.8961744905 | German shepherd (0.90); malinois (0.01); kelpie (0.00); dingo (0.00); soccer ball (0.00) | TRUE |
| german_shepherd_4.jpg | german_shepherd | 231 | collie | 0.2355108112 | collie (0.24); kelpie (0.19); Tibetan mastiff (0.13); German shepherd (0.09); Border collie (0.04) | FALSE |
| hummingbird_1.jpg | hummingbird | 94 | hummingbird | 0.9061354399 | hummingbird (0.91); window screen (0.00); jacamar (0.00); redshank | TRUE |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | (0.00); admiral (0.00) | |
| hummingbird_2.jpg | hummingbird | 94 | hummingbird | 0.9095668793 | hummingbird (0.91); water ouzel (0.00); jacamar (0.00); window screen (0.00); hip (0.00) | TRUE |
| hummingbird_3.jpg | hummingbird | 917 | comic book | 0.1504553556 | comic book (0.15); vine snake (0.07); toucan (0.06); walking stick (0.06); green mamba (0.05) | FALSE |
| hummingbird_4.jpg | hummingbird | 94 | hummingbird | 0.8628011942 | hummingbird (0.86); jacamar (0.01); indigo bunting (0.01); water ouzel (0.00); European gallinule (0.00) | TRUE |
| jellyfish_1.jpg | jellyfish | 107 | jellyfish | 0.9013634324 | jellyfish (0.90); matchstick (0.00); missile | TRUE |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | (0.00); balloon (0.00); volcano (0.00) | |
| jellyfish_2.jpg | jellyfish | 107 | jellyfish | 0.9013041258 | jellyfish (0.90); isopod (0.00); sea slug (0.00); nematode (0.00); sea anemone (0.00) | TRUE |
| jellyfish_3.jpg | jellyfish | 107 | jellyfish | 0.9048047662 | jellyfish (0.90); coral reef (0.00); brain coral (0.00); spotted salamander (0.00); balloon (0.00) | TRUE |
| jellyfish_4.jpg | jellyfish | 329 | sea cucumber | 0.7695545554 | sea cucumber (0.77); jellyfish (0.10); sea urchin (0.01); flatworm (0.01); chiton (0.00) | FALSE |
| tarantula_1.jpg | tarantula | 885 | velvet | 0.295746088 | velvet (0.30); jellyfish (0.12); volcano (0.05); | FALSE |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | balloon (0.04); parachute (0.03) | |
| tarantula_2.jpg | tarantula | 980 | volcano | 0.1908383071 | volcano (0.19); matchstick (0.10); spider web (0.08); jellyfish (0.05); goldfish (0.05) | FALSE |
| tarantula_3.jpg | tarantula | 76 | tarantula | 0.8517101407 | tarantula (0.85); barn spider (0.03); tick (0.01); wolf spider (0.00); black widow (0.00) | TRUE |
| tarantula_4.jpg | tarantula | 506 | coil | 0.05781359226 | coil (0.06); volcano (0.04); shovel (0.03); leatherback turtle (0.03); wok (0.03) | FALSE |
| tiger_1.jpg | tiger | 292 | tiger | 0.7085394263 | tiger (0.71); tiger cat (0.14); lynx (0.00); Norwegian elkhound (0.00); jaguar (0.00) | TRUE |

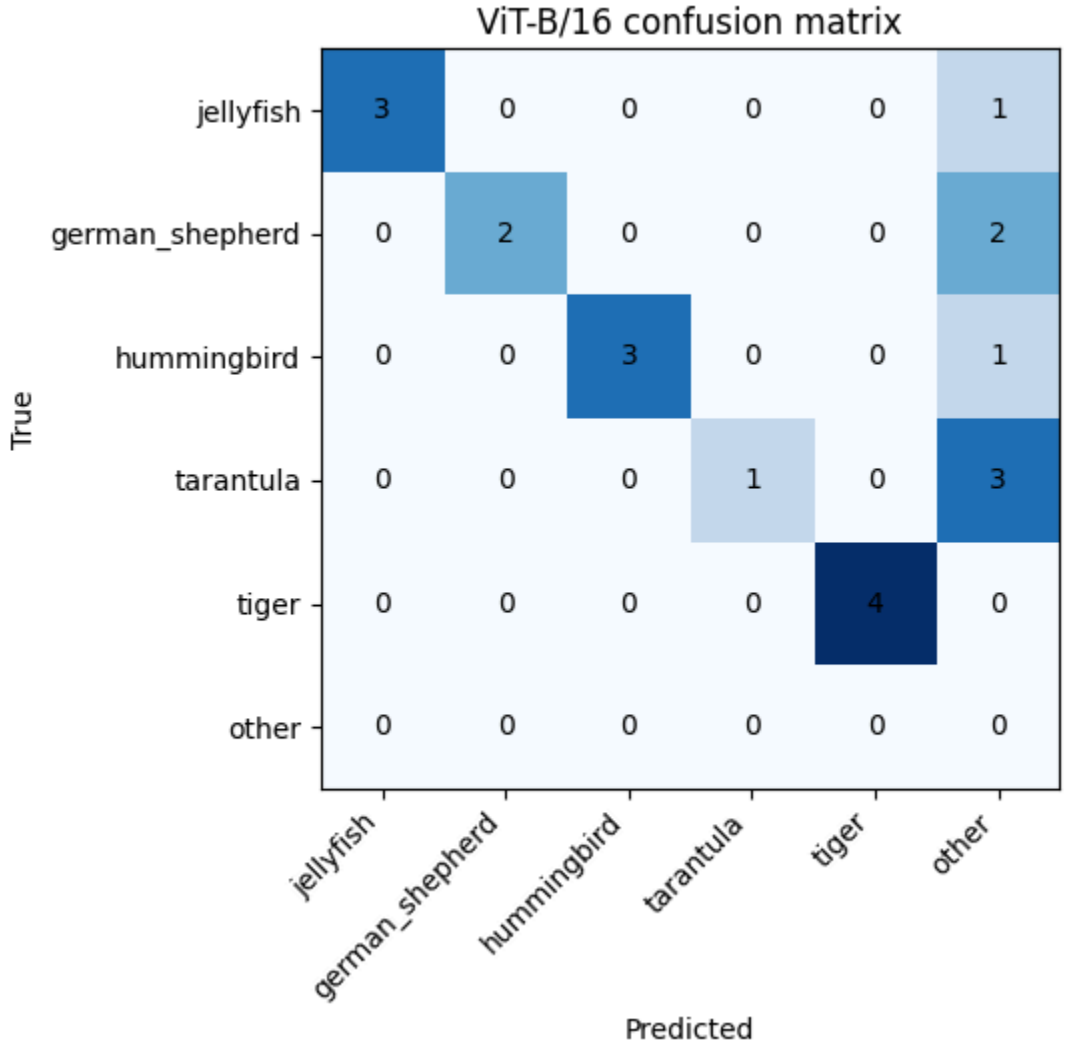
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| | | | | | | |
|---|---|---|---|---|---|---|
| tiger_2.jpg | tiger | 292 | tiger | 0.6516503692 | tiger (0.65); tiger cat (0.20); lynx (0.00); jaguar (0.00); airship (0.00) | TRUE |
| tiger_3.jpg | tiger | 292 | tiger | 0.8050143123 | tiger (0.81); tiger cat (0.09); jaguar (0.01); lynx (0.00); zebra (0.00) | TRUE |
| tiger_4.jpg | tiger | 292 | tiger | 0.614367187 | tiger (0.61); tiger cat (0.17); jaguar (0.00); Gila monster (0.00); gondola (0.00) | TRUE |

Overall top-one accuracy - 0.650 (13/20)

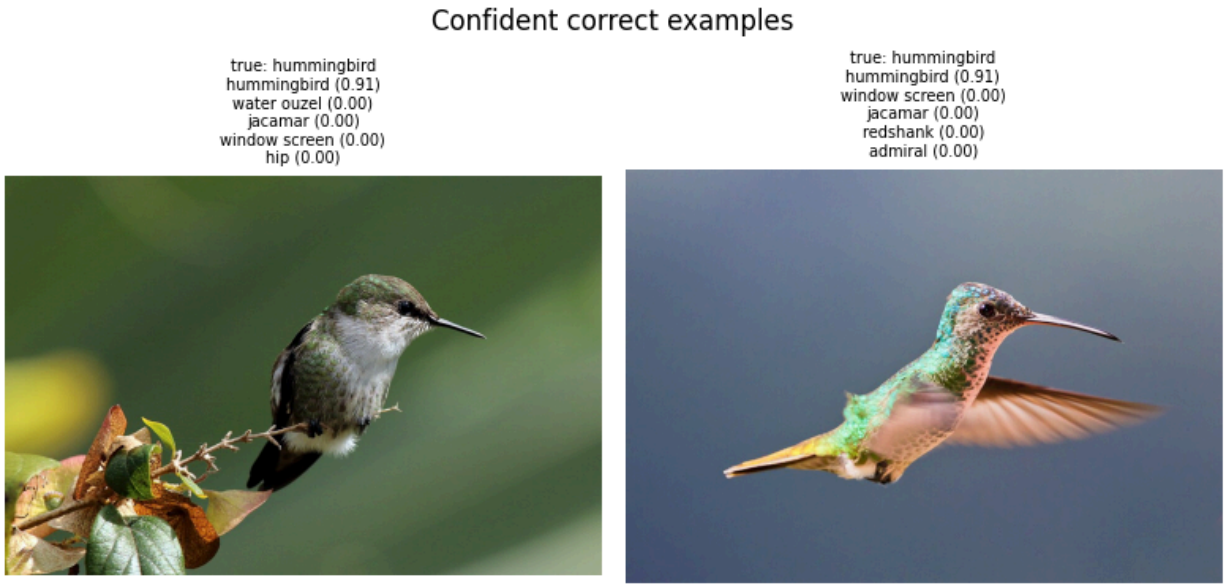
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Confusion Matrix:



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Two correct Examples:




Note: These images performed very well and had high confidence scores (0.91). These are some of the more clearly conceptually evident images that I selected, so they scored very well.

Three Unsuccessful Examples:




Note: These three unsuccessful examples are primarily because of bad image selection, as images were selected randomly from Wikimedia Commons' database. For example, the german shepherd was likely identified as a dinosaur because of the fact that it's a skeleton, and not a living dog. Images which were more conceptually clear performed much better, and it would be best to consider these incorrect images as anomalies.

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**Fixed Output Limitation:**

The model's output layer has 'only' 1000 categories, so any image examples which are outside of these 1000 classes will immediately be mislabelled. The model is forced to classify each image, so mislabelling is an inevitability. These images which sit outside the output layer might be easily identified by poor classification scores, like the three above unsuccessful examples. The model is essentially matching an image's makeup to a label; if the actual label doesn't exist in the model's output layer, the classification is incorrect, which is why models with higher categorical libraries always perform better at image classification.

Image Sources (images were pulled from Wikimedia Commons):

| file | category | title | author | license | url |
|---|---|---|---|---|---|
| jellyfish_1.jpg | jellyfish | File:Lion's mane jellyfish in Gullmarn fjord at Sämstad 8.jpg | W.carter | CC0 | https://commons.wikimedia.org/wiki/File:Lion%27s_mane_jellyfish_in_Gullmarn_fjord_at_S%C3%A4mstad_8.jpg |
| jellyfish_2.jpg | jellyfish | File:Jellyfish from Corpus Christi.jpg | Miguel Angel Omaña Rojas | CC0 | https://commons.wikimedia.org/wiki/File:Jellyfish_from_Corpus_Christi.jpg |
| jellyfish_3.jpg | jellyfish | File:Moon jellyfish at Gota Sagher.JPG | Alexander Vasenin | CC BY-SA 3.0 | https://commons.wikimedia.org/wiki/File:Moon_jellyfish_at_Gota_Sagher.JPG |
| jellyfish_4.jpg | jellyfish | File:Jellyfish at rathmullan beach.jpg | Mary McGuire | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:Jellyfish_at_rathmullan_beach.jpg |
| german_shepherd_1.jpg | german_shepherd | File:German Shepherd dog. "Canis lupus familiaris".jpg | Museum of Veterinary Anatomy FMVZ USP / Wagner Souza e Silva edited by Rodrigo Tetsuo Argenton | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:German_Shepherd_dog._%E2%80%9CCanis_lupus_familiaris%E2%80%9D.jpg |

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| | | | | | |
|---|---|---|---|---|---|
| german_shepherd_2.jpg | german_shepherd | File:20110425 German Shepherd Dog 8505.jpg | Jakub Hałun | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:20110425_German_Shepherd_Dog_8505.jpg |
| german_shepherd_3.jpg | german_shepherd | File:German shepherd enjoying the sun.jpg | PROPOLI87 | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:German_shepherd_enjoying_the_sun.jpg |
| german_shepherd_4.jpg | german_shepherd | File:Stray German Shepherd Dog.jpg | Satdeep Gill | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:Stray_German_Shepherd_Dog.jpg |
| hummingbird_1.jpg | hummingbird | File:Colibrí Cola de Oro (Golden-tailed Sapphire Hummingbird) Bigger File.jpg | Marcial4 | CC BY-SA 3.0 | https://commons.wikimedia.org/wiki/File:Colibr%C3%AD_Cola_de_Oro_(Golden-tailed_Sapphire_Hummingbird)_Bigger_File.jpg |
| hummingbird_2.jpg | hummingbird | File:Vervain hummingbird (Mellisuga minima).jpg | Charles J. Sharp | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:Vervain_hummingbird_(Mellisuga_minima).jpg |
| hummingbird_3.jpg | hummingbird | File:Haeckel Trochilidae.jpg | Ernst Haeckel | Public domain | https://commons.wikimedia.org/wiki/File:Haeckel_Trochilidae.jpg |
| hummingbird_4.jpg | hummingbird | File:Bee hummingbird (Mellisuga helenae) immature male.jpg | Charles J. Sharp | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:Bee_hummingbird_(Mellisuga_helenae)_immature_male.jpg |
| tarantula_1.jpg | tarantula | File:A New View of the Tarantula Nebula.jpg | NASA | Public domain | https://commons.wikimedia.org/wiki/File:A_New_View_of_the_Tarantula_Nebul |

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| | | | | | |
|---|---|---|---|---|---|
| | | | | | a.jpg |
| tarantula_2.jpg | tarantula | File:Tarantula Nebula by JWST.jpg | NASA, ESA, CSA, STScI, Webb ERO Production Team | Public domain | https://commons.wikimedia.org/wiki/File:Tarantula_Nebula_by_JWST.jpg |
| tarantula_3.jpg | tarantula | File:Brachypelma edit.jpg | User:Fir0002 | CC BY-SA 3.0 | https://commons.wikimedia.org/wiki/File:Brachypelma_edit.jpg |
| tarantula_4.jpg | tarantula | File:30 Doradus, Tarantula Nebula.jpg | NASA, ESA, ESO, D. Lennon and E. Sabbi (ESA/STScI), J. Anderson, S. E. de Mink, R. van der Marel, T. Sohn, and N. Walborn (STScI), N. Bastian (Excellence Cluster, Munich), L. Bedin (INAF, Padua), E. Bressert (ESO), P. Crowther (Sheffield), A. de Koter (Amsterdam), C. Evans (UKATC/STFC, Edinburgh), A. Herrero (IAC, Tenerife), N. Langer (AifA, Bonn), I. Platais (JHU) and H. Sana (Amsterdam) | Public domain | https://commons.wikimedia.org/wiki/File:30_Doradus,_Tarantula_Nebula.jpg |
| tiger_1.jpg | tiger | File:Panthera tigris altaica 13 - Buffalo Zoo.jpg | Dave Pape | Public domain | https://commons.wikimedia.org/wiki/File:Panthera_tigris_altaica_13_-_Buffalo_Zoo.jpg |
| tiger_2.jpg | tiger | File:Amur Tiger Panthera tigris altaica Eye 2112px edit.jpg | Photo (c)2007 Derek and Julie Ramsey (Ram-Man) edit by Chris_huh. | Public domain | https://commons.wikimedia.org/wiki/File:Amur_Tiger_Panthera_tigris_altaica_Ey |

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| | | | | | |
|---|---|---|---|---|---|
| | | | | | e_2112px_edit.jpg |
| tiger_3.jpg | tiger | File:Bengal tiger (Panthera tigris tigris) female 3.jpg | Charles J. Sharp | CC BY-SA 4.0 | https://commons.wikimedia.org/wiki/File:Bengal_tiger_(Panthera_tigris_tigris)_female_3.jpg |
| tiger_4.jpg | tiger | File:Tiger-2.jpg | Hans Stieglitz | CC BY-SA 3.0 | https://commons.wikimedia.org/wiki/File:Tiger-2.jpg |

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# PART 2

## Task 1:

My Logits vs. model's built-in output:

```
[Task 1] max |my logits - built-in logits| = 1.72e-05 | allclose: True
[Task 1] accuracy with 'T2: a photo of a {category}': 1.000
```

My logits matched the model's very closely. A difference of only 1.72e-05 means that the difference between my logits and those of the model were extremely small, and functionally negligible.

Example on tiger_1.jpg:

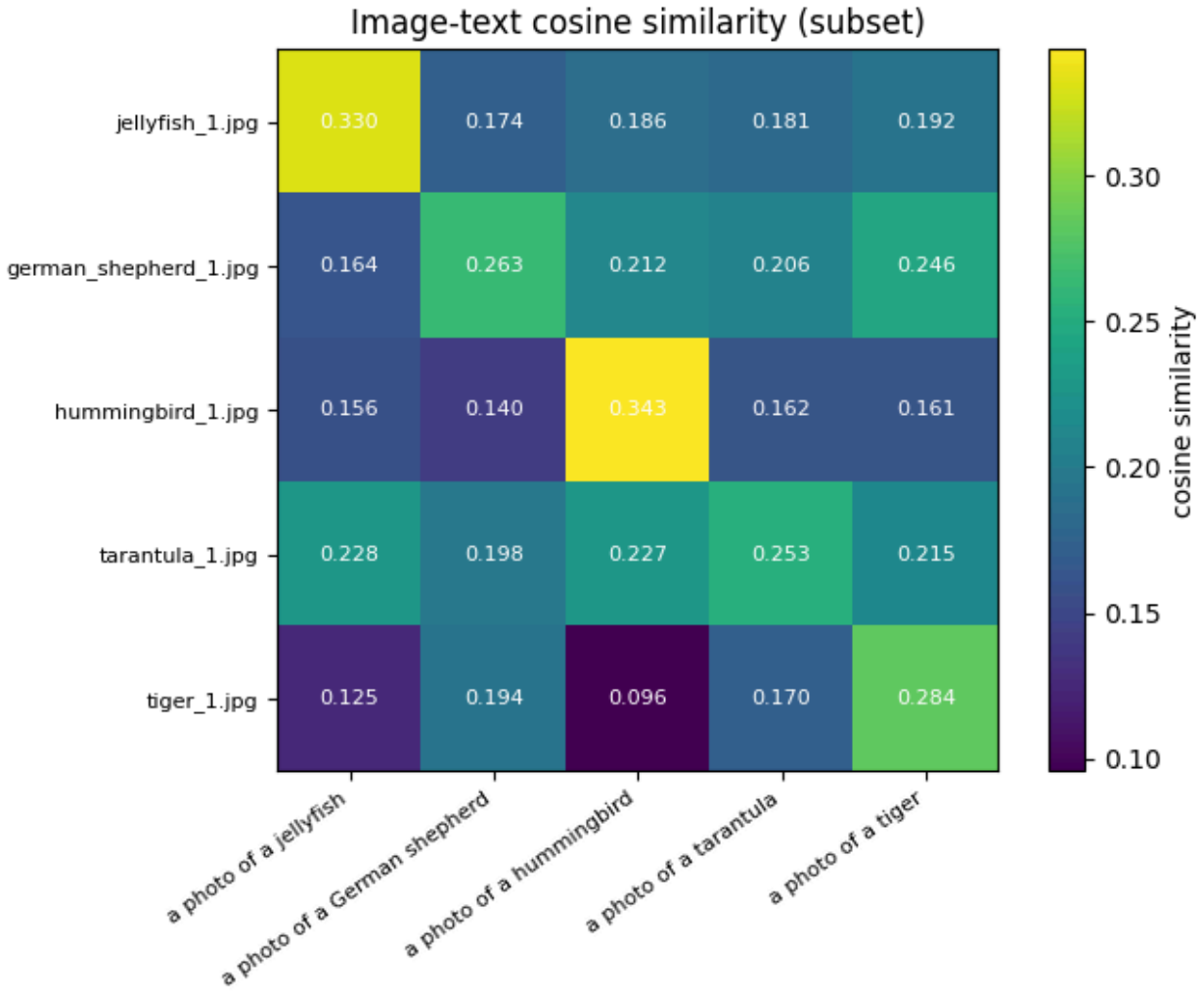
| | prompt | cosine | logit | softmax |
|---|---|---|---|---|
| 0 | a photo of a jellyfish | 0.1252 | 12.520000 | 0.0000 |
| 1 | a photo of a German shepherd | 0.1942 | 19.420000 | 0.0001 |
| 2 | a photo of a hummingbird | 0.0959 | 9.590000 | 0.0000 |
| 3 | a photo of a tarantula | 0.1703 | 17.030001 | 0.0000 |
| 4 | a photo of a tiger | 0.2836 | 28.360001 | 0.9999 |

Candidate-set demo on tiger_1.jpg:

- cosine with 'a photo of a tiger' 5 candidates: 0.2836 | 6 candidates: 0.2836
- softmax prob of 'a photo of a tiger' 5 candidates: 0.9999 | 6 candidates (+tiger cat): 0.4378

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Image-text Similarity Heatmap:




**Comparison of computed logits with the model's built-in output:**

Cosine similarity is the dot product of two vectors, which measures the relationship or similarity between image and text embeddings. A higher cosine similarity generally signals that an image and its related textual embedding or description are closely related, and therefore that CLIPs textual output is correct.

Logits are those cosines which are multiplied in order to widen the gap (i.e. 0.284 -> 28.4) and further stratify the model's cosine scores, which allows for a more fine-tuned output. The ordering of the scores is kept, but this prepares the scores for softmax-ing.

The softmax function converts the raw logits into a probability distribution, so that all probabilities add up to 1 are probabilities which are closer to 1 are more likely to be a correct textual embedding of the image.

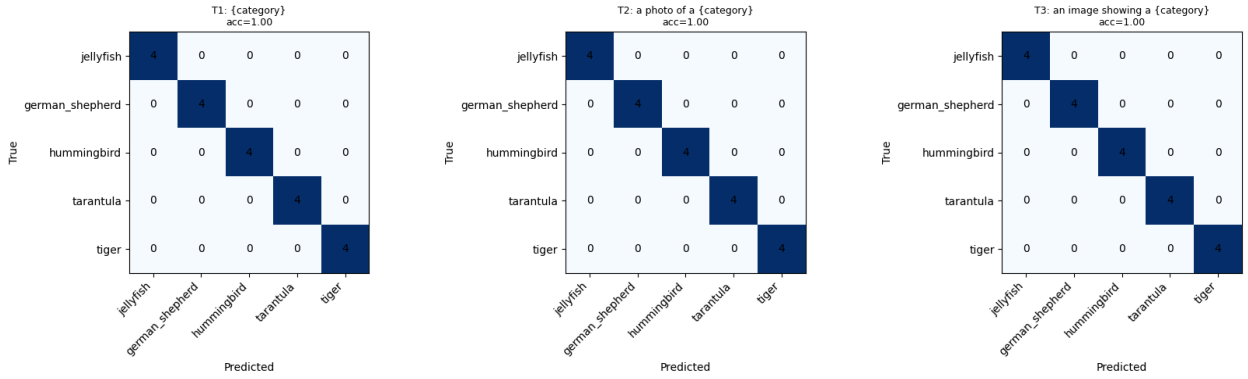
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**Dependence on comparison texts being compared:**

Softmax divides by the sum of all candidates, so it doesn't necessarily say whether a text embedding is true, just that it is more likely. Every text embedding is scored relative to other embeddings, so that a softmax score is just a probability that a candidate is more likely than another. In the demonstration with 'tiger_1.jpg,' the softmax scoring dropped from 0.9999 for tiger to 0.4378 once a distractor 'tiger cat' was added. A single candidate was able to drop the probability by that much even though the logit score / cosine similarity remained the same.

# Task 2:

Confusion matrix for each template:




Images with the largest change in class score after wording:

| | file | true | pred T1 | P(true) T1 | pred T2 | P(true) T2 | pred T3 | P(true) T3 | P(true) range | pred changes |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | german_shepherd_1.jpg | german_shepherd | german_shepherd | 0.803 | german_shepherd | 0.840 | german_shepherd | 0.601 | 0.239 | False |
| 12 | tarantula_1.jpg | tarantula | tarantula | 0.975 | tarantula | 0.849 | tarantula | 0.772 | 0.203 | False |
| 13 | tarantula_2.jpg | tarantula | tarantula | 0.970 | tarantula | 0.952 | tarantula | 0.899 | 0.071 | False |
| 3 | german_shepherd_4.jpg | german_shepherd | german_shepherd | 0.978 | german_shepherd | 0.963 | german_shepherd | 0.916 | 0.062 | False |
| 15 | tarantula_4.jpg | tarantula | tarantula | 0.993 | tarantula | 0.975 | tarantula | 0.944 | 0.050 | False |
| 11 | jellyfish_4.jpg | jellyfish | jellyfish | 0.999 | jellyfish | 1.000 | jellyfish | 0.999 | 0.001 | False |

**Why changing a prompt differs from fine-tuning:**

Changing a prompt changes only the text going into the text encoder, where fine-tuning updates weights based on labeled data, which permanently changes how the model connects inputs to embeddings (this is how the model 'learns').

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**Comparison of results with Part 1:**

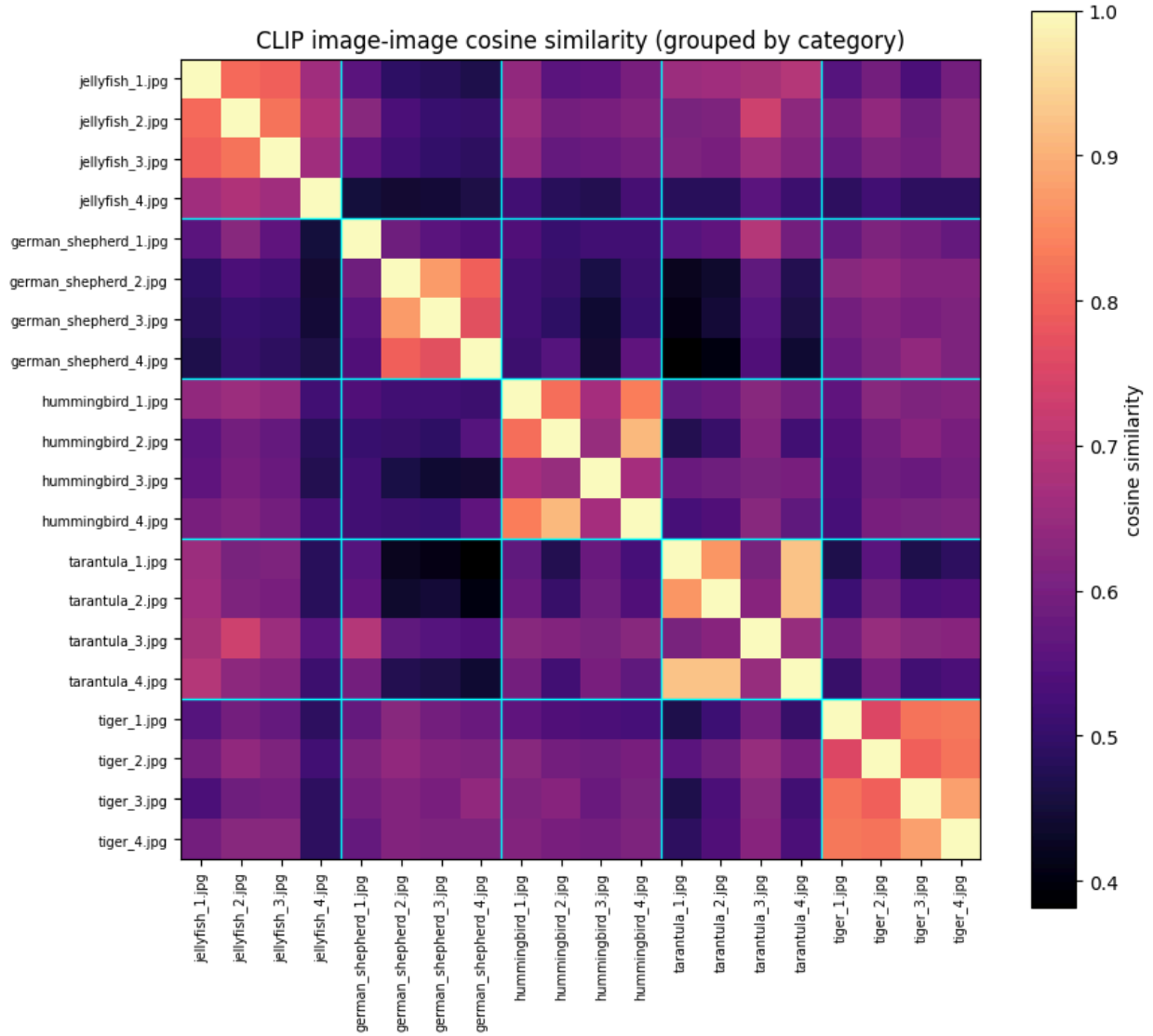
| true | ViT correct | CLIP correct T1 | CLIP correct T2 | CLIP correct T3 |
|---|---|---|---|---|
| german_shepherd | 0.50 | 1.0 | 1.0 | 1.0 |
| hummingbird | 0.75 | 1.0 | 1.0 | 1.0 |
| jellyfish | 0.75 | 1.0 | 1.0 | 1.0 |
| tarantula | 0.25 | 1.0 | 1.0 | 1.0 |
| tiger | 1.00 | 1.0 | 1.0 | 1.0 |

CLIP was able to correctly classify 20/20 images, while ViT was only able to correctly classify 13/20. ImageNet's method of classification is based on labelled data, whereas CLIP was trained using image-text scrapers to collect image-text pairs. The ViT chose only among 1000 classes, while CLIP was basically given a five-way choice, making it so that the probability of being wrong was much lower. The poor ViT scoring for german_shepherd makes sense, as two of the images were low-quality, but the poor performance of ViT when identifying tarantulas is surprising.

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# Task 3:

Image-Image Cosine Similarity Matrix:




```
[Task 3] mean similarity within category: 0.754 | across categories: 0.557
[Task 3] nearest neighbour is in the same category for 18/20 images
```

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Most similar image pairs within categories:

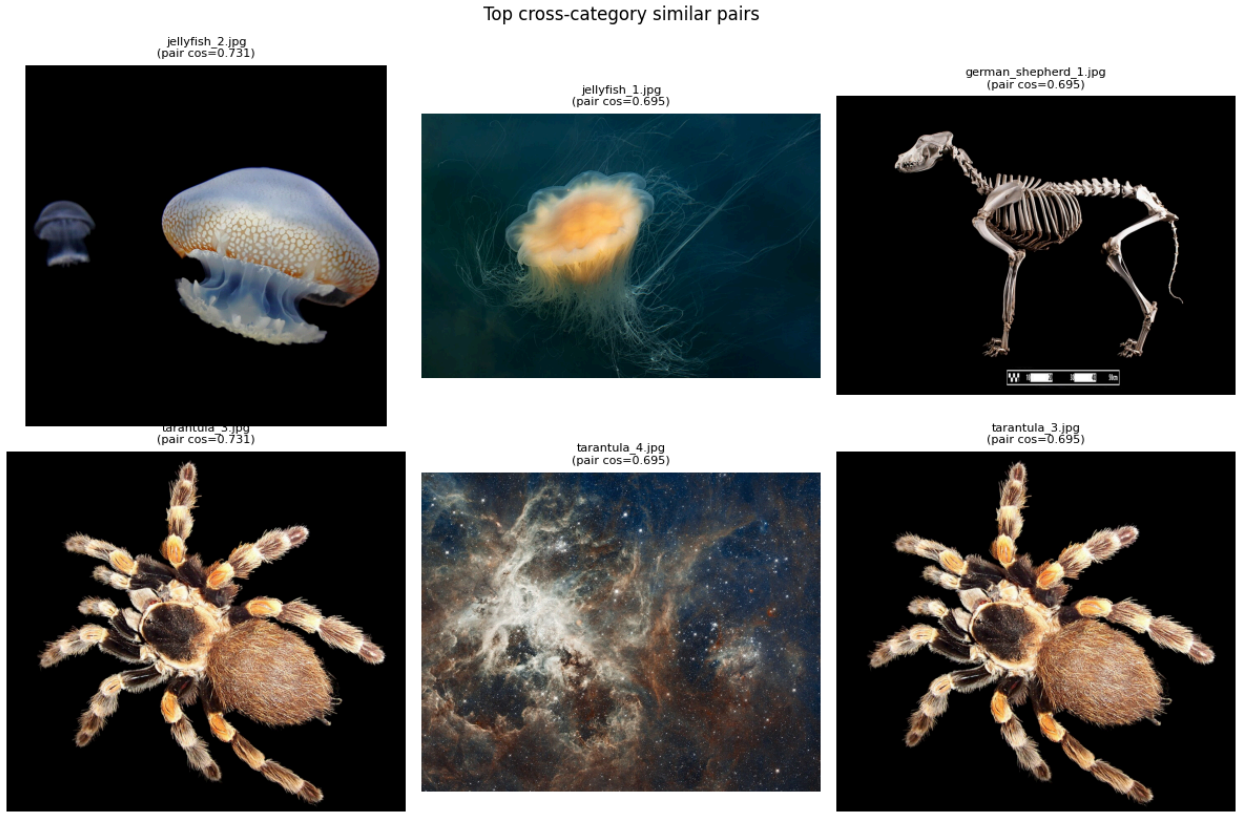
```
Most similar pair inside each category:
  jellyfish        jellyfish_2.jpg <-> jellyfish_3.jpg  (0.821)
  german_shepherd  german_shepherd_2.jpg <-> german_shepherd_3.jpg  (0.876)
  hummingbird      hummingbird_2.jpg <-> hummingbird_4.jpg  (0.912)
  tarantula        tarantula_2.jpg <-> tarantula_4.jpg  (0.929)
  tiger            tiger_3.jpg <-> tiger_4.jpg  (0.878)

Most similar pairs ACROSS categories:
  jellyfish_2.jpg <-> tarantula_3.jpg  (0.731)
  jellyfish_1.jpg <-> tarantula_4.jpg  (0.695)
  german_shepherd_1.jpg <-> tarantula_3.jpg  (0.695)
  jellyfish_1.jpg <-> tarantula_3.jpg  (0.673)
  jellyfish_1.jpg <-> tarantula_2.jpg  (0.660)
```

`Least similar pair WITHIN a category: german_shepherd_1.jpg <-> german_shepherd_4.jpg (0.543)`

*This least-similar pairing was likely skewed by the strange image selection for german_shepherd_1.jpg, which was a skeleton.

Top cross-category similar pairs:



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**Similarities based on visual properties:**

The similarities are generally based on the background. The jellyfish + tarantula and german-shepherd + tarantula pairs are present most likely because of the stark black background, which makes the images look quite similar regardless of the actual subject of the image.

**What CLIP's representation captures:**

Generally, CLIPs representation is most dependent on the actual image subject in most cases, as images with matching subjects are much more likely to have similar image embeddings (which seems obvious). Interestingly though, it's fairly obvious that while CLIPs representation primarily (and hopefully) uses the subject matter as a primary identifying feature, the representation is also dependent upon other meta-features of the image, which in my examples were the black backgrounds in some of the image pairs.